

10/21/25

- From last class: Weak law of large numbers says that

$$P(|Y_n - \mu| > \epsilon) < \delta \quad \text{for all } n > N \quad (\delta = \text{delta})$$
 where $\mu = E[X]$, $Y_1 = X_1$, $Y_n = \frac{1}{n}(X_1 + \dots + X_n)$
- by Chebyshev: $P(|Y_n - \underbrace{E[Y_n]}_{\mu}| > \epsilon) < \text{Var}(Y_n)/\epsilon^2 \rightarrow \text{Var}(X)/n \cdot \epsilon^2 < \delta$

- we want to simulate $E[X]$, knowing that $E[X]$ is $\text{Bern}(0.32)$.
 How many trials to guarantee approximation within 0.05 with prob 0.99? Using Chebyshev.
 Bernoulli $\text{Var}(X) = p(1-p) = 0.32(0.68) = 0.2176$
 $\epsilon = 0.05$ $\delta = (1 - 0.99) = 0.01$
 $n = 0.2176 / (0.01 \cdot 0.05^2) = 8704$

- $N = \text{random var}$, $S = X_1 + X_2 + \dots + X_N$

$$E[S] = E[S|N=1] \cdot P(N=1) + \dots + E[S|N=n] \cdot P(N=n) \rightarrow \sum_{n=1}^{\infty} E[S|N=n] \cdot P(N=n)$$

$$= \sum_{n=1}^{\infty} E[X_1 + \dots + X_n] \cdot P(N=n) = \sum_{n=1}^{\infty} n \cdot E[X] \cdot P(N=n)$$

$$= E[X] \sum_{n=1}^{\infty} n \cdot P(N=n) = E[X] \cdot E[N]$$

Not on midterm
From here on

Continuous Probability

- imagine the spinner wheel from Twister - it has infinite points it could stop at, all within $[0, 1)$. $P(X \in [0, 1/4]) = 1/4$. $P(X \in [1/4, 3/4]) = 1/2$
 $P(X \in [a, b]) = b - a$

If $b=a \rightarrow P(X \in [a, a]) = 0 = P(X=a)$

- unintuitive, but: in 1d sample space, 0d objects (points) have 0 probability.
 - same is true of any dimension sample space: for an event to exist, it must be on the same dimensional level as the space itself.

- to find $E[X]$, we've been using prob. mass function. However, this is broken ($P(X=a) \approx 0$)
 instead, we use prob. density function: $f_X(x) = \frac{d}{dx} F_X(x)$
 can also use cumulative distribution function (CDF): $F_X(x) = P(X \leq x)$